

Yuchen Wang

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Education

University of Illinois Urbana-Champaign , Computer Science	Champaign, IL, USA 2025 – 2027
Peking University , Intelligent Science and Technology (Artificial Intelligence) • Zhi Class (Elite Program), Top 10%	Beijing, China 2021 – 2025

Experience

Alibaba U.S. , Research Intern — LLM Training Algorithms & Infrastructure	Sunnyvale, CA, USA May 2026 – Dec 2026 8 months
- Core contributor to Occamy-1.0 (35B-A3B); co-designed staged SFT/HDPO, model soup, and SAO with trainable context compaction and cross-segment credit assignment. 500K+ downloads across Occamy variants. - Architected environment-/capability-first data synthesis and multi-harness training infrastructure with state-aware graders and asynchronous actor-critic learning; the deduplicated SFT union spans 15K trajectories and 403M tokens. - Shipped Occamy with Accio; versus its Qwen3.6 base, the released model raised AutomationBench strict pass from 7.5% to 27.6% and Claw-Eval average from 69.5 to 82.2, while reducing Claw-Eval T/C full-trial time by 36.6%.	
Freedo Technology , Research Intern — 3D Vision and Generative AI	Beijing, China May 2025 – Aug 2025 4 months
Built an end-to-end depth-conditioned ControlNet pipeline for 3D reconstruction from noisy point clouds, improving geometric fidelity from 32.7% to 85.3%.	

Open Source Projects

Open-Source LLM Systems Engineering

RL infrastructure, long-context training, inference optimization, optimizer correctness, and training-data pipelines across NVIDIA NeMo, Megatron-LM, vLLM, SGLang, FlashInfer, and Hugging Face.

- RL infrastructure: bypassed the Ray driver for [NeMo RL teacher top-k transfers](#), achieving 4.44–5.31× speedup in controlled 8×B200 transfer measurements. Corrected async GRPO/distillation checkpoint-resume bookkeeping in [TRL](#) focused CPU checks passed, with end-to-end training-resume validation pending.
- Long-context training: developing [Qwen4-Exp integration in Megatron Bridge](#) for QSA, per-layer embeddings, gated residuals, and HF conversion; GPU integration remains pending. Added [fused GDN Q/K normalization](#) and [selective Mamba recompute](#) in Megatron-LM. Made packed QSA selection document-local in [mcore-bridge](#), 2.70× faster on an 8K synthetic selector case.
- Inference memory and correctness: in-place BF16 MoE conversion in [vLLM](#) halved 120B TP2 conversion peak allocation from 7.88 to 3.94 GiB; fixed calibrated FP8 KV prefill in [FlashInfer](#). Fixed Mamba cache-publication ordering in [SGLang](#).
- Optimizer correctness: fixed small-norm normalization in [Muon](#) and implemented checkpoint-safe Adam counters in [Megatron-LM](#). CPU/DCP round-trip checks passed; full CUDA training validation remains pending.
- Training data: preserved source sharding in [HF Datasets streaming shuffle](#), restoring all four DataLoader workers on 19 source shards, with deterministic epoch and checkpoint-restoration coverage.

Projects

RareDx — Graph-Grounded RL for Rare-Disease Diagnosis

Co-first author; advised by Prof. Tianyu Liu at Tsinghua University. Graph-grounded rewards for partial diagnostic matches.

- Co-designed rewards using disease-ontology distance, biomedical similarity, and phenotype-disease links. With SFT and retrieval, 27B diagnosis Hit@10 increased from 28.53% to 40.76%.

CineFlow — Dependency-Driven Parallel Video Generation

Co-first author; advised by Prof. Fan Lai at UIUC. Dependency-driven video diffusion using semantic DAGs, runtime trajectory fusion, online edge pruning, and critical-path scheduling without retraining.

- Analyzed 1K+ MovieGenBench prompts; more than 68% exhibit non-linear dependencies.
- Across Wan2.2-5B, CogVideoX-5B, and HunyuanVideo on 8×H100, achieved 1.7–5.5× end-to-end speedup, 1.30–2.02× lower P90 latency, and 5.4–17.3% higher VBench overall.

Dynamic Prefill Optimization via Adaptive Online Packing

AIMD control with live p95 TTFT feedback and greedy/DP prompt packing.

- Achieved up to 20% lower TTFT on production-style DynamoLLM traces.

FlashAttention-style CUDA Optimization for GPT-2

Tiled online softmax, memory-efficient attention, and kernel fusion.

- Reduced HBM traffic by roughly 10× and achieved up to 9% end-to-end speedup.

Reinforcement Learning for Legal Reasoning

Advised by Prof. Yansong Feng at Peking University; Zero-RL → distilled-CoT SFT → GRPO for structured legal multiple-choice reasoning.

- Reached 57.6% accuracy, outperforming SFT-only and RL-only baselines.

Academic Service

- Reviewer, [WSDM 2027](#).

Awards

Merit Student

Zhi Class Scholarship

CMO (Provincial First Prize), 2020

Skills

- Python, C++, CUDA, PyTorch, Ray, FSDP, Megatron-LM, NeMo RL, vLLM, SGLang, Nsight, RL, distributed training, diffusion transformers, ControlNet, 3D vision, and point clouds